

The Informational Bifurcation of Hidden Liquidity: Aggressive versus Midpoint Concealment on a Single Venue

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Abstract

Hidden liquidity is commonly treated as a single object carrying a single informativeness. Using event-by-event LOBSTER message data for 474 NASDAQ-traded names over 2023–2024, we show it is neither, and that the second half of that claim is the more consequential. Hidden executions away from the midpoint are associated with a positive signed markout while executions *at* the midpoint carry a small negative one, so any classifier that force-signs midpoint fills averages an informed population against an uninformed one and returns approximately zero. That split survives a matched tick-rule placebo and is the within-venue signature of the self-selection equilibrium of Zhu (2014). Its *level*, however, is not robust: the same footprint measures +2.09 bps under the conventions most commonly adopted, +0.62 bps when prints are signed only where the aggressor is unambiguous and no burst is formed, and zero when burst formation is additionally denied sight of price. We report +0.62 as the conservative level, show that almost all of it is impounded within one second of the print, and argue that the convention-dependence itself—not the magnitude—is what the evidence supports.

1 Introduction

That hidden orders carry information is established (Bessembinder et al., 2009; Hautsch and Huang, 2012). That informed traders sort themselves across execution mechanisms according to their demand for immediacy is likewise established, in theory by Zhu (2014) and empirically across trading venues by Comerton-Forde and Putniņš (2015) and Buti et al. (2017). This paper joins the two. It asks whether the sorting is visible *inside* a single venue and a single order type, and finds that it is—sharply enough that the informativeness of hidden liquidity cannot be summarized by one number.

The empirical object is the type-5 print in the NASDAQ ITCH message stream: an execution against non-displayed resting liquidity. Roughly half of these prints execute exactly at the prevailing midpoint and half away from it. The two halves behave differently in a way that is not a matter of degree. Prints away from the midpoint are associated with a signed three-minute midpoint move that is flat across every horizon we measure—out to thirty minutes, and, against a time-of-day-stratified placebo, all the way to the closing bell on the full 470-name panel. Prints at the midpoint carry -0.42 bps at three minutes, and more negative values at longer horizons than the placebo of §4.3 shows to be partly artifact. Because the standard trade-sign classifiers differ chiefly in how they handle midpoint trades, they disagree on the aggregate footprint by two full basis points—from +2.09 under a rule that abstains on them to a statistical zero under the Ellis–Michael–O’Hara hybrid (Ellis et al., 2000), which force-signs them. That disagreement is not a robustness failure to be adjudicated. It is the bifurcation showing through the classifier.

Our contribution is threefold, and the third is a negative result about our own measurement instrument.

First, we measure the sorting within a venue. A cross-venue comparison confounds self-selection with venue-level differences in fee structure, clientele, and reporting latency. Here, venue, order type, and reporting mechanism are held fixed, and the only variable is whether the execution crossed the spread. We regard this as a sharper test of the Zhu (2014) mechanism than the cross-venue evidence, and we are explicit that the mechanism itself is not our discovery.

Second, we characterize the aggressive footprint as the adverse-selection component of the spread. The evidence is a set of cross-sectional facts rather than a structural estimate: the footprint is a stable fraction of the quoted half-spread across a fivefold range of spread widths; volatility, which predicts it univariately, contributes nothing once the spread is controlled; and on the Huang and Stoll (1997) accounting it is between 22% and 51% of the effective half-spread, depending on whether price movement coincident with a touch sweep is counted as information or as displacement (§6.3); it is incremental to visible order-flow imbalance; and it survives the removal of information-event windows. A direct test of mechanical displacement is only partly favourable, and we report it as such. We are careful to distinguish the facts from the interpretation. A stable ratio is consistent with the Glosten–Milgrom mechanism (Glosten and Milgrom, 1985) under which the spread compensates providers for informed flow; it does not demonstrate that the spread is *set* that way, and we do not test the causal direction.

Third, we show what our own headline measurement overstates. Conditioning on aggressive same-side execution may select moments at which the midpoint is already travelling in that direction, in which case a forward markout measures continuation rather than response. We test this directly with an event-time profile on both sides of the print, restricted to prints executing outside the prevailing quote so that the aggressor’s side cannot be misassigned. Nearly half the signed move—47.5% over $[-30s, +180s]$ —precedes the print. A Hasbrouck VAR with genuine memory at the horizons of interest, which conditions on the prior return path, independently finds the response decaying to zero or below by ten minutes. We withdraw the causal reading accordingly, and report the association. We then test the mechanism that would explain both facts—that the markout is continuation of a move already underway—and find that it does not hold: regressing the forward markout on the pre-print drift leaves essentially all of it orthogonal to the prior path, with the continuation component economically zero and slightly negative. Half the move precedes the print, but the pre-drift does not forecast what follows, and the markout–VAR disagreement therefore remains open rather than resolved.

Section 2 describes the data and the burst construction. Section 3 reports the negative result that motivates the rest: on two pilot names, clustering *visible* aggressive flow yields no informed footprint under any parameterization we tried or under two alternative reconstructions. That section is pilot-scale throughout and we mark it as such. Section 4 establishes the bifurcation. Section 5 characterizes the aggressive footprint as an adverse-selection quantity. Section 6 tests the two leading non-informational accounts. Section 7 reports what unconditional markouts get wrong. Section 9 states what this design cannot settle.

2 Data and Construction

2.1 Data

The analysis uses event-by-event limit-order-book data from LOBSTER, reconstructed directly from the NASDAQ ITCH message stream, over 2023–2024. The panel is 474 names with sufficient hidden flow, spanning 237,948 raw ticker-days, of which roughly 40% are absent from the staged

archive or fail book reconstruction. Each test then applies the filter its estimator requires, so analysis-ready name-day counts differ across tables and are reported in each caption. No name-day is excluded on the basis of its outcome.

Two properties of the data govern what can be claimed.

LOBSTER does not disclose the side of hidden executions. The Direction field is uniformly +1 for type-5 prints, so hidden trades carry no native sign and must be classified. This is not incidental to the paper; it is the reason the bifurcation is visible at all, since it forces an explicit decision about midpoint prints that a signed feed would obscure.

LOBSTER reconstructs the NASDAQ book only. All midpoints, quotes and depths are NASDAQ’s, not the consolidated NBBO. Section 9 states the consequence, which is material.

Because each ticker-day must be streamed from the raw archive, decompressed and reconstructed individually, we restrict to a two-year window rather than a longer one.

2.2 Burst construction

Isolated order submissions are noisy at horizons beyond a few seconds, so we aggregate temporally clustered same-side executions into discrete *bursts*. For burst b with start time t_b and reference midpoint m_{t_b} , the signed aggregate volume is $Q_b = \sum_{j \in b} \text{side}_j q_j$.

Cluster membership is determined by a *self-exciting counter*, which we describe precisely because its status is often overstated in this literature. A running intensity decays exponentially between events and is incremented by each arrival:

$$\lambda(t_k) = \lambda(t_{k-1}) e^{-\beta(t_k - t_{k-1})} + 1, \quad \lambda(t_0) = 1, \quad (1)$$

and the burst remains active while λ exceeds a threshold $\lambda_{\min}$. This is the intensity recursion of an exponential-kernel Hawkes process, and we use it as a *clustering heuristic*; we do not fit it. No likelihood is maximized, no branching ratio is estimated, and no goodness-of-fit is reported, so we do not describe the construction as a fitted point process and none of our claims rest on its being one. What we do verify is that the clustering it detects is not an arrival-rate artifact: the count of same-side clustered bursts exceeds an *inhomogeneous*-Poisson null matched to each name’s empirical intraday intensity profile by a median z of 47, rejecting the Poisson null on 100% of name-days.

Equation (1) has two free parameters, not three. The excitation increment is fixed at unity, which is a normalization rather than a choice: scaling the increment and $\lambda_{\min}$ together leaves the detector’s behaviour unchanged, so only their ratio is identified. The decay rate β and the threshold $\lambda_{\min}$ are the genuine parameters, and both are varied— $\beta \in \{0.5, 1, 2, 5\}$ and $\lambda_{\min} \in \{0.3, 0.5, 0.8\}$ —in Section 3. Additional filters on fractional sweep volume, directional consistency, and opposing-volume ratio are applied uniformly and described in the replication materials.

Bursts of *hidden* executions, which carry the paper’s results, are formed identically on type-5 prints alone.

2.3 Inference

Throughout, the day—not the individual print—is the unit of statistical inference. Markouts are equal-weighted within name-day, then averaged across the cross-section, and t -statistics are Newey–West (Newey and West, 1987) with 10 lags on the resulting daily mean series, or day-level Fama–MacBeth (Fama and MacBeth, 1973) with the same correction where a regression is reported. An outlier rule is applied before any figure is quoted: name-days whose markout exceeds 1,000 bps in absolute value at any horizon are dropped. On the primary panel this removes 56 name-days of 221,261 (0.03%); one reaches $-114,627$ bps, which is a corrupt price rather than a return.

3 A Two-Name Null for Clustered Visible Flow

We begin with the negative result that directs the paper toward hidden liquidity, and state its scope before its content: everything in this section is measured on two pilot names, AAPL and TSLA. It is what motivated the reconstruction choice the rest of the paper makes, not a cross-sectional claim, and we do not extend it to one.

Clustering *visible* aggressive executions produces no measurable informed footprint, and this does not depend on how the clustering is tuned. Table 1 recomputes the three-minute directional markout over the decay grid $\beta \in \{0.5, 1, 2, 5\}$; every cell is economically negligible and well inside the bid–ask spread, and minimum cluster size (over $\{3, 5, 10\}$, omitted) has no effect. The threshold $\lambda_{\min}$ was varied over $\{0.3, 0.5, 0.8\}$ during detector calibration with the same result. The null is not an artifact of a particular parameterization.

3-min markout (bps)	$\beta = 0.5$	$\beta = 1$	$\beta = 2$	$\beta = 5$
AAPL	+0.02	−0.09	−0.03	+0.14
TSLA	−0.10	+0.07	−0.02	+0.26

Table 1: Decay-rate grid for the visible-flow clustering rule, evaluated ungated. Every configuration is sub-spread. Minimum cluster size $\in \{3, 5, 10\}$ is omitted as it had no effect.

Because tuning does not help, we replace the construction entirely with three alternative reconstructions of the same message stream: *order-flow imbalance* (OFI; the top-of-book queue dynamics of Cont et al., 2014, and its deep-learning descendants, Kolm et al., 2023), *book resilience* (aggressive sweeps whose consumed depth fails to replenish), and *hidden-execution* clustering. Table 2 reports the comparison. Only the hidden-execution reconstruction produces a positive markout on both names.

3-min markout (bps)	Clustered visible	OFI	Book resilience	Hidden
AAPL	−0.08	+0.08	+0.19	+0.42
TSLA	+0.08	+0.08	+0.16	+0.31

Table 2: Four reconstructions of the same message stream, evaluated ungated on the 2023 sample. Hidden bursts are signed by the quote rule (execution price against the prevailing midpoint); the other three use aggregated signed flow. Two names only: OFI and book resilience were validated at pilot scale, and we make no cross-sectional claim for them.

Two construction choices are dictated by the definitions and applied uniformly. The book-resilience non-refill condition is defined on queue *depth* rather than quote price, and its markout is measured strictly after the depth-observation window closes, so the filter cannot select on the path over which it is subsequently evaluated. And hidden prints, being unsigned in the feed, are signed by the quote rule throughout.

The rest of the paper concerns the one reconstruction that survives. The scope limit bears repeating because the sections that follow are cross-sectional and this one is not: all four reconstructions were compared on two names, so this is the reason we study hidden-execution clustering, not a ranking of the four at cross-sectional scale. A referee is entitled to read Section 3 as motivation rather than evidence, and nothing downstream depends on it.

4 The Bifurcation

4.1 Classifier disagreement

The headline hidden-execution footprint is measured with the quote rule, which is undefined for the 48.9% of type-5 prints executing exactly at the midpoint and therefore abstains on them. Since that convention discards half the data, the natural question is whether the footprint survives classifiers that instead *assign* a sign to midpoint prints. Table 3 re-signs under four canonical rules.

Classifier (3-min hidden markout)	Markout (bps)	NW t
Quote rule, abstaining at the midpoint	+2.09	+25.3
Tick rule (midpoint prints signed by last price change)	+0.62	+9.5
CLNV (Chakrabarty et al., 2007)	+0.46	+8.3
EMO (Ellis et al., 2000)	+0.03	+0.6

Table 3: Three-minute hidden-execution footprint under four trade-sign classifiers (474 names, 2023–2024, 199,627 name-days; 50% of prints at the midpoint). The aggregate ranges over two basis points depending solely on the treatment of midpoint prints.

The disagreement is not diffuse. It is concentrated entirely in names whose hidden prints are predominantly at the midpoint. Sorting names into quartiles by at-midpoint share (Table 4), all four classifiers agree where few prints are at the midpoint (Q1, 15% share: +0.74 to +1.80); where most are (Q4, 91%), only the abstaining rule retains a positive value while every force-signing rule goes to zero or negative.

At-mid quartile	At-mid share	Abstain	Tick	EMO	CLNV
Q1 (fewest at-mid)	15%	+1.80	+1.47	+0.74	+1.28
Q2	45%	+1.42	+0.69	+0.19	+0.45
Q3	75%	+1.98	+0.18	−0.31	+0.09
Q4 (most at-mid)	91%	+2.74	−0.27	−0.52	−0.31

Table 4: Three-minute footprint (bps) by classifier across quartiles of a name’s at-midpoint print share. Classifier disagreement appears only as the at-midpoint share rises, locating it in the treatment of midpoint prints rather than in the underlying measurement. The Q4 row carries a composition caveat given in the text.

The Q4 cell requires a caveat that we state rather than bury, because it is the cell that most favours our preferred convention. The abstaining rule’s dependent variable is estimated *only* on non-midpoint prints, so the sorting variable mechanically governs the size of the sample on which the outcome is measured. Across quartiles the number of signed prints per name-day falls from 436 in Q1 to 34 in Q4—a factor of thirteen. The +2.74 in Q4 is therefore estimated from roughly 9% of those names’ hidden prints, and is correspondingly noisier and potentially selected: abstention may be isolating the most informative subsample rather than measuring a population. This does not affect the within-name-day bifurcation of §4.2, which does not sort on this variable, but it does mean the quartile gradient should not be read as a clean cross-sectional relationship.

4.2 Two populations

Table 5 measures the two populations separately, and is the paper’s central result. Hidden executions away from the midpoint are associated with a +2.09 bps three-minute move that is flat across horizons (+2.09 $\rightarrow$ +2.24 $\rightarrow$ +2.25 over 3/15/30 minutes). Hidden executions at the midpoint carry a small *negative* markout (−0.42 at three minutes; the more negative values at fifteen and thirty minutes are discounted in §4.3, where the placebo recovers 15–16% of them against 5.4% at three minutes), which is the signature of uninformed liquidity provision absorbing transient price moves.

Hidden-execution subset	3 min	15 min	30 min
Aggressive (away from the midpoint)	+2.09 ($t=25.3$)	+2.24 ($t=12.2$)	+2.25 ($t=13.3$)
At the midpoint (tick-signed)	−0.42 ($t=−6.6$)	−0.64 ($t=−7.0$)	−0.71 ($t=−10.7$)

Table 5: Signed midpoint markout (bps) of aggressive versus at-midpoint hidden executions (474 names, 2023–2024, 199,627 name-days). Pooling the two is what produces the lower, classifier-dependent aggregate of Table 3.

We flag a definitional caution the reader should carry forward. Prints away from the midpoint are, by construction, prints with an identifiable aggressor, so the statement “aggressor-signed prints carry adverse selection and unsigned prints do not” is partly a property of trade classification rather than a purely economic finding. What is *not* definitional is the sign of the second population. A classification artifact would place at-midpoint prints near zero; they instead carry a reliably negative markout that grows with horizon.

That defence is only as good as the sign it rests on, and the at-midpoint row is tick-signed. The tick rule labels a print a buy precisely when the price last ticked up, so if short-horizon reversion follows that preceding tick, a negative forward markout would arise mechanically for *any* set of moments signed this way—no liquidity-provision economics required, and the growth with horizon would be equally expected. Since this is the load-bearing evidence that the bifurcation is economic rather than classificational, we test it directly in §4.3 rather than argue it.

The reading is that concealment is informationally bifurcated. Concealed *aggressive* intent is associated with adverse selection; concealed *passive* midpoint provision is not. This refines rather than restates the finding that hidden orders are informative: only the aggressive half is, and reporting a single pooled number averages an informed population against an uninformed one.

4.3 The at-midpoint markout is not a tick-rule artifact

The concern of the preceding paragraph has a sharp implication: if tick-rule conditioning manufactures negative markouts, it should do so wherever it is applied, including at moments that are not hidden midpoint executions at all. We test it two ways on the same panel.

A matched placebo. We take the at-midpoint print times of each name-day and shift them all by a single random per-day offset. Inter-arrival structure—and therefore the burst clustering, which is spacing-dependent—is preserved exactly, while the moments themselves are decoupled from actual hidden midpoint executions. We then read the midpoint at those shifted times, sign it by the identical tick rule, cluster identically, and measure the same markouts. If the negative row is a property of tick-signing rather than of midpoint concealment, the placebo should reproduce it.

Tick-conditioning a known-informed population. We re-sign the *aggressive* prints by the pure tick rule, discarding their quote-rule sign. This asks what tick conditioning does to a

population we have independent reason to believe is informed.

Tick-rule placebo	3 min	15 min	30 min
At-midpoint prints (tick-signed)	−0.411 ($t=-9.0$)	−0.587 ($t=-10.4$)	−0.674 ($t=-11.1$)
Matched placebo (shifted times, tick-signed)	−0.022 ($t=-1.2$)	−0.088 ($t=-3.3$)	−0.110 ($t=-2.9$)
<i>Tick-conditioning the aggressive leg (3 min)</i>			
Aggressive, quote-rule signed	+1.232 ($t=+26.5$)		
Aggressive, pure tick-rule signed	−0.064 ($t=-1.6$)		

Table 6: Tick-rule placebo tests (470 names, 2023–2024, 235,940 name-days). The placebo recovers 5.4% of the at-midpoint markout at three minutes and is statistically zero; tick-signing the aggressive population destroys its footprint rather than reversing it. Levels in the lower panel are estimated on this run’s filter (which additionally requires 180s of within-session history and a valid pre-print quote) and are therefore not directly comparable to Table 5’s +2.09; the informative quantity is the *within-panel* contrast between the two rows.

Neither test supports the artifact reading. The placebo earns −0.022 bps at three minutes against the at-midpoint row’s −0.411—5.4% of it, and statistically indistinguishable from zero. At fifteen and thirty minutes the placebo does become significant, but recovers only 15% and 16% of the corresponding at-midpoint values, so tick-rule conditioning contributes a detectable but small negative drift at longer horizons and essentially none at the three-minute horizon the paper’s tables use.

The second test is the more informative of the two. Re-signing the aggressive prints by the pure tick rule does not turn their footprint negative; it drives it to zero (−0.064, $t = -1.6$, against +1.232 under the quote rule). What tick conditioning does is *destroy* directional information, not manufacture the appearance of contrarian trading. Both results point the same way: the negative markout on at-midpoint hidden prints is a property of those executions, not of the classifier applied to them.

We state the residual limitation. These tests establish that the tick rule does not manufacture the sign; they do not establish that it recovers the *correct* sign for any individual midpoint print, and given the second row above—where tick-signing a known population yields nothing—the tick rule is evidently a weak instrument. The at-midpoint markout should therefore be read as evidence that this population differs sharply from the aggressive one, which is what the bifurcation claims, rather than as a precise estimate of uninformed liquidity provision. Signing midpoint prints against a consolidated quote, where a print at the NASDAQ midpoint may sit away from the NBBO midpoint and acquire a genuine sign, would settle it; see §9.

4.4 The bifurcation is not an averaging artifact

A pooled term structure that is flat could in principle blend continuation in some names against reversion in others. Table 7 splits the aggressive-hidden term structure by price quintile. It does not blend: the markout is flat or gently building in every quintile and reverts in none. It is larger in cheap, tick-constrained names (+3.96 bps in Q1 against +1.42 in Q5), which Section 5 shows is the spread effect.

Price quintile	Avg. price	3 min	15 min	30 min
Q1 (tick-constrained)	\$14	+3.96	+4.21	+4.19
Q2	\$34	+1.89	+2.62	+2.61
Q3	\$58	+1.53	+1.51	+1.46
Q4	\$97	+1.53	+1.58	+1.43
Q5 (large-price)	\$281	+1.42	+1.43	+1.46

Table 7: Aggressive-hidden markout (bps) by price quintile (474 names, 2023–2024, $\sim 44,000$ name-days per quintile; all $t > 12$). Flat or building in every quintile, reverting in none.

4.5 The intraday term structure, on a subsample

The horizons above stop at thirty minutes. To instrument the rest of the session we extend the aggressive-hidden markout to one hour, two hours and the burst-to-close midpoint, and net each horizon against a *time-of-day-stratified* placebo that draws each burst’s counterfactual time within its own 30-minute intraday bucket, so the null preserves the U-shaped intensity and drift profile a uniform placebo would ignore.

Aggressive-hidden markout	Burst (bps)	TOD placebo (bps)	Net (bps)	Net t (day)
3 min	+2.023	+0.007	+2.017	+28.3
15 min	+2.122	−0.022	+2.145	+14.8
30 min	+2.166	−0.101	+2.268	+15.5
1 hour	+2.131	−0.145	+2.283	+15.3
2 hour	+2.076	−0.289	+2.378	+10.3
to close	+2.014	−0.415	+2.433	+10.0

Table 8: Intraday term structure of the aggressive-hidden footprint on the *full panel* (470 names, 2023–2024, $\approx 206,070$ name-days after the outlier rule). Each horizon is netted against a time-of-day-stratified placebo drawing each burst’s counterfactual time within its own 30-minute bucket; t -statistics are Newey–West on the daily-mean net series. The *gross* markout is flat across the whole session—+2.02 at three minutes, +2.01 measured to the close—so the profile does not depend on the placebo to stay level.

The profile is flat across the entire session on both measures. The gross markout runs +2.02, +2.12, +2.17, +2.13, +2.08, +2.01 from three minutes to the close, and the placebo is small at every horizon ($|\cdot| \leq 0.42$), so the net rises only gently from +2.02 to +2.43. A hidden burst’s directional association with price is established within roughly three minutes and then neither grows nor decays for the remaining six and a half hours.

One structural caveat attaches to the final row and we state it rather than bury it. Every other horizon measures a fixed elapsed time, whereas “to close” measures to a fixed endpoint, so a counterfactual timestamp drawn earlier in its bucket has a systematically longer window than the burst it replaces; this is why the placebo drifts monotonically more negative with horizon. In the earlier 48-name version of this table that mattered, because the gross to-close markout there fell to +1.42 and the row netted to +2.08 only because the placebo was simultaneously more negative. On the full panel it does not: the gross markout at the close is +2.014, statistically indistinguishable from its three-minute value, so the flat profile holds without any placebo adjustment at all.

4.6 The sign is not a stale-midpoint artifact

Because trades are signed against a reconstructed midpoint, a print above a *lagging* midpoint could be mechanically classified a buy just before the midpoint ticks up, manufacturing a positive markout. Table 9 re-signs three ways.

Signing rule (3-min aggressive-hidden markout)	bps	NW t
Baseline (midpoint at the trade)	+2.00	+23.8
Staler midpoint (quote 1s earlier)	−0.02	−0.4
Forward midpoint (quote 1s later; look-ahead)	−1.59	−39.2
Outside-the-quote prints only (unambiguous sign)	+2.09	+12.5

Table 9: Signing robustness (474 names, 2023–2024, 146,393 name-days). Signing against a staler midpoint degrades rather than inflates the footprint, the opposite of the stale-midpoint prediction; and on prints executing outside the quote, where the sign cannot be wrong, the footprint is undiminished.

Signing against a deliberately staler midpoint does not inflate the footprint—it collapses it to zero, because a staler midpoint misclassifies more prints, which is the opposite of what the stale-midpoint hypothesis predicts. Most directly, restricting to prints executing *outside* the prevailing quote, where the aggressor’s side is unambiguous regardless of any midpoint error, leaves the footprint at +2.09 bps ($t = 12.5$). The measurement is strongest exactly where the sign is most certain.

We record here, and return to in Section 7, that this outside-the-quote subset is also the one on which the pre-print evidence is cleanest—and there it cuts the other way.

5 The Footprint Is an Adverse-Selection Quantity

5.1 It scales with the spread, and not with volatility

If the aggressive footprint is the adverse-selection component of the spread, it carries a sharp cross-sectional prediction: it should be a roughly constant *fraction* of the quoted half-spread across names, and should track the spread specifically rather than volatility, illiquidity or price level generally. This is a stronger test than anything that precedes it, because a mechanical or volatility-scaling account does not predict it.

The prediction holds. Across 469 names with at least 100 name-days, the correlation between the quoted half-spread and the permanent price-impact component is +0.85, and the ratio of the two is centered on 0.34 with an interquartile range of 0.28–0.42. Sorting into quintiles by quoted half-spread, the spread ranges over a factor of five (2.47 to 12.54 bps) while the adverse-selection *share* moves only between 0.29 and 0.38.

Column (3) is the discriminating test. Realized volatility predicts the footprint strongly on its own (+0.573, $t = 15.0$). Entered jointly with the quoted half-spread it collapses to +0.008 ($t = 0.2$) while the spread coefficient is if anything larger. The footprint is not a volatility phenomenon that happens to appear in wide-spread names; it is a spread phenomenon for which volatility was proxying.

One caveat constrains this. Wide-spread names have a coarser effective price grid, so a given midpoint move is mechanically larger there, and that channel inflates measured impact in wide-spread names in a way we cannot fully separate from the informational one. It does not, however,

Regressor (standardized)	(1)	(2)	(3)
Quoted half-spread	+0.850 (+34.8)		+0.930 (+25.2)
Realized volatility		+0.573 (+15.0)	+0.008 (+0.2)
log price			−0.042 (−0.9)
log dollar volume			+0.224 (+4.8)
R^2	0.723	0.328	0.755

Table 10: Cross-name determinants of the permanent price-impact component (465 names, 2023–2024; all variables standardized, t -statistics in parentheses). Univariately both the quoted half-spread and realized volatility predict the footprint. Jointly, the spread coefficient is undiminished while volatility falls to zero.

explain the *stability of the ratio* across quintiles, nor the disappearance of the volatility coefficient under the spread control, since tick coarseness and volatility are themselves positively related.

5.2 Its share of trading costs

Table 11 places the footprint on the Huang and Stoll (1997) accounting, estimated on the aggressive-hidden trades with $\tau = 180$ seconds.

Huang–Stoll component (aggressive-hidden trades)	bps	NW t
Quoted half-spread	+6.39	+27.4
Effective half-spread (taker pays)	+4.00	+23.5
= Price impact (adverse selection)	+2.02	+32.4
+ Realized half-spread (provider retains)	+1.97	+16.2

Table 11: Effective/realized spread decomposition (Huang and Stoll, 1997) for aggressive-hidden trades (474 names, 2023–2024, 203,142 name-days). Reported for the *ratio* it yields, not as independent confirmation—see the text.

We are explicit about what this does and does not add, because it is easy to overstate. The Huang–Stoll price-impact component is defined as $D_t(M_{t+\tau} - M_t)$, which is *algebraically the markout itself*; the realized half-spread is then the effective half-spread minus that quantity. The near-coincidence of the +2.02 bps component with the +2.09 bps markout of Table 5 is therefore not independent corroboration. It is the same estimand computed twice on slightly different name-day filters, and any claim that the decomposition *confirms* the markout would be circular.

What the decomposition genuinely contributes is the ratio, which the markout alone cannot supply: the permanent component is 2.02 of a 4.00 bps effective half-spread, or 51% of what the taker actually pays, and 32% of the 6.39 bps quoted half-spread. Roughly half the effective cost of aggressive hidden execution is compensation for information and roughly half is the provider’s gross margin.

5.3 The adverse-selection share is a range, not a number

That 51% is measured from the print, and §6.3 shows that most of the level so measured coincides with the touch being swept within 100ms. Whether that movement belongs in the numerator is exactly the question the sweep test could not settle, so the share must be reported as a range. We re-estimate the decomposition separately on swept and unswept prints, and with the price-impact leg measured from $t + 1s$ so that anything coincident with the touch changing is excluded.

	Quoted/2	Effective/2	Impact from t	From $t + 1s$	Share
Swept prints	2.744	1.845	2.785	0.359	151% / 19.5%
Unswept prints	2.747	1.708	0.397	0.381	23.2% / 22.3%

Table 12: Huang–Stoll decomposition by sweep group (470 names, 2023–2024, bps). “Share” gives price impact as a fraction of the effective half-spread, measured from t and from $t + 1s$ respectively. Levels here are estimated under this run’s filter and are not directly comparable to Table 11; the informative content is the *within-run* contrast between groups and between measurement origins.

Three things are worth reading off this table. First, the two groups face *indistinguishable quoted spreads* (2.744 against 2.747), so the difference between them is not a compensation difference—it is what happens after the print. Second, the swept group’s impact measured from t is 151% of the effective half-spread, which is not a possible adverse-selection share: it implies a negative realized half-spread, the provider losing more than they charge. That figure is displacement being counted as information, and its impossibility is the clearest evidence that the from- t measurement overstates. Third, once the first second is excluded the two groups converge to 19.5% and 22.3%—the same quantity, as one would expect if what remains is a common informational component.

We therefore report the adverse-selection share of the effective half-spread as **between roughly 22% and 51%**: 51% if movement coincident with the touch changing is price discovery, 22% if it is displacement, with the truth in between given that §6.4 shows maker withdrawal is involved in 43% of sweeps. The qualitative claim—that a substantial share of the effective cost of aggressive hidden execution is compensation for information rather than provider margin—holds across the whole range. The point estimate does not.

5.4 It is incremental to visible order-flow imbalance

Hidden flow might simply proxy for the visible order-flow imbalance already known to forecast short-horizon returns (Cont et al., 2014), in which case the footprint would be a re-description rather than an addition. We bin each name-day on a fixed ten-second clock and regress the forward three-minute midpoint return on quote-rule-signed aggressive-hidden volume and on the visible Cont–Kukanov–Stoikov imbalance, both standardized within name-day, univariately and jointly.

The hidden-flow coefficient is not attenuated by the control; it *rises*, from +0.160 to +0.198 bps per SD, with the t -statistic nearly doubling. This requires explanation, because two positively correlated regressors that both positively predict the outcome should attenuate one another. Enhancement instead indicates classical suppression, which arises when the conditional correlation between regressors is negative—and here it is. The mechanism is specific to this pair: a hidden *buy* consumes displayed depth at the ask, and ask-side depletion enters the Cont–Kukanov–Stoikov imbalance with a *negative* sign, so the two carry mildly opposing contemporaneous components even though both forecast returns positively.

The magnitude reconciles exactly, which is worth showing because a correlation this small looks

Predictor of forward 3-min midpoint return	bps per SD	NW t
<i>Univariate</i>		
Aggressive-hidden signed volume	+0.160	+12.2
Visible OFI (Cont et al., 2014)	+2.351	+54.6
<i>Joint</i>		
Aggressive-hidden signed volume	+0.198	+22.9
Visible OFI	+2.330	+54.2

Table 13: Hidden flow versus visible order-flow imbalance (474 names, 2023–2024, 168,556 name-days, ten-second bins; day-level Fama–MacBeth with Newey–West errors). Controlling for visible OFI does not attenuate the hidden-flow coefficient.

too weak to move a coefficient by a quarter. For standardized regressors, $b_H^{\text{joint}} = (b_H^{\text{uni}} - \rho b_O^{\text{uni}})/(1 - \rho^2)$, and substituting the measured values gives $(0.1601 - (-0.0161)(2.3508))/(1 - 0.0161^2) = 0.1980$ against the 0.1981 reported. The suppression term is 24% of b_H^{uni} precisely because b_O^{uni} is 14.7 times larger: a small correlation multiplied by a much larger coefficient is a large adjustment *relative to the smaller one*. The implied ρ of -0.016 also matches the directly measured median name-day correlation of -0.027 on the same clock.

Visible OFI is by far the larger short-horizon predictor in absolute terms, an order of magnitude above the hidden coefficient, and we do not claim otherwise. The point is only that the hidden footprint survives it intact.

6 Two Non-Informational Accounts

6.1 Information events

The most natural alternative to an adverse-selection reading is that aggressive hidden executions cluster around information events—earnings, guidance, macro releases—and that price moves because of the announcement, with the execution a correlate. We obtain quarterly earnings dates from Yahoo Finance, retrieved through the `yfinance` interface, for the 418 panel names that report (4,925 announcements over 2023–2024; the remaining 56 are ETFs, which do not). We proxy for *unscheduled* news by the magnitude of the name’s daily return. We note that Yahoo Finance is a retail-grade source for announcement timestamps; a commercial calendar would be preferable, and the dates enter only as an exclusion window rather than as a treatment variable.

The confound is not supported. Excluding earnings windows moves the footprint from +1.910 to +1.874; excluding the largest decile of daily moves leaves +1.751; excluding both, retaining 88% of the panel, leaves +1.760 against +1.878, or 94% retention with the t -statistic essentially unchanged. The same pattern holds on the Huang–Stoll series ($+2.02 \rightarrow +1.99$).

The event cells are themselves informative rather than a nuisance: the footprint roughly doubles around earnings (+3.00) and rises monotonically with the size of the daily move (+2.96, +3.43, +3.98 at the top decile, ventile and percentile). That is what adverse selection predicts when more private information is in play, and it is difficult to reconcile with a purely mechanical account of the same quantity.

We note the limit. Return magnitude is a coarse proxy for news: it captures unscheduled information only insofar as it moved prices that day, and cannot distinguish news arrival from a liquidity shock. A timestamped intraday news feed would be the full treatment. What the

Sample	Footprint (bps)	NW t	name-days
All name-days (earnings-covered names)	+1.910	+29.3	154,577
Earnings window (± 1 day)	+3.002	+13.9	7,657
Excluding earnings window	+1.874	+29.6	146,920
Excluding earnings (± 3 days)	+1.858	+31.4	141,715
Largest 10% of daily moves	+2.959	+27.1	17,559
Excluding largest 10% of moves	+1.751	+23.6	143,527
Excluding earnings ± 2d and top 5% moves	+1.760	+26.8	143,111

Table 14: The footprint after removing information-event windows (474-name panel, 2023–2024; three-minute quote-rule markout). On the joint-clean subsample—no earnings window, no top-decile move, 88% of all name-days—the footprint retains 94% of its value.

test establishes is that the scheduled-announcement channel accounts for essentially none of the measured footprint.

6.2 Mechanical quote displacement

The most serious competing account of the footprint is mechanical rather than informational. An aggressive execution consumes displayed depth; if the level is depleted the best quote moves, the midpoint moves with it, and if the level is never replenished that displacement persists. Such an account reproduces every stylised fact above—a footprint appearing instantly, never reverting, scaling with the spread, largest in coarse-tick names—without any information being conveyed.

Testing it requires conditioning on something known strictly *before* the print, and the natural candidate—the ratio of trade size to displayed depth at the touch—turns out not to be the right quantity. A type-5 print executes against non-displayed liquidity, so when the hidden order rests inside the spread the print does not consume the displayed queue at all. That ratio measures order size relative to visible liquidity, not the fraction of the queue consumed, and a print that is small on that measure may still be followed microseconds later by the same aggressor’s marketable remainder sweeping the touch. The mechanical channel that actually applies to hidden prints is that sequential sweep, and we test it directly below.

6.3 The sequential-sweep channel

We therefore observe the sweep directly. For each aggressive hidden print we record the touch on the side being taken immediately before the print and again 100ms later, and call the level *swept* if its price moved away or its size more than halved at an unchanged price. The channel is common: 29.8% of aggressive hidden prints are followed by a sweep within 100ms.

The measurement requires care, because the obvious test is circular. Conditioning on the touch moving inside the first second and then measuring a markout that begins at the print conditions on most of the outcome—the footprint is largely impounded within that first second, and for a buy a rising ask *is* a rising midpoint. We therefore separate the conditioning window from the measurement window: prints are classified on $[t, t + 100\text{ms}]$ alone, and the forward move is measured from $t + 1\text{s}$ and from $t + 5\text{s}$, strictly after the displacement. Table 15 reports both the circular and the clean versions so the contamination is visible rather than asserted.

Two findings, and the second is uncomfortable for the reading advanced above.

Markout window	Swept	Unswept	Gap
From $t + 1\text{s}$ to $t + 180\text{s}$	+0.374 (9.7)	+0.439 (22.0)	−0.066
From $t + 5\text{s}$ to $t + 185\text{s}$	+0.203 (5.8)	+0.260 (15.1)	−0.057

Table 15: Forward move by whether the touch was swept within 100ms of the print (470 names, 2023–2024, 235,940 name-days; bps, NW t in parentheses; median 153 swept and 378 unswept prints per name-day). Both rows separate the conditioning window from the measurement window; a version measuring from t itself would condition on a quote move inside the interval it then measures, and is not reported for that reason.

The sweep does not mark differentially informative flow. Measured from after the displacement, swept and unswept prints are indistinguishable, and if anything swept prints carry slightly *less* forward move (+0.374 against +0.439 from $t + 1\text{s}$; +0.203 against +0.260 from $t + 5\text{s}$). Whatever a sweep signals, it does not predict a larger subsequent revision. The apparent $18\times$ concentration is entirely an artifact of the overlap between conditioning and measurement.

But the level of the measured footprint is concentrated in the first second. The clean rows show that once the displacement window is excluded, swept and unswept prints carry between +0.20 and +0.44 bps, against an all-print average measured from the print itself of roughly +1.3. Most of what a from-the-print markout records therefore occurs while the touch is changing, and we do not claim that displacement can be excluded as its source.

6.4 What actually happens at the touch: consumption versus withdrawal

Calling all of that “displacement” assumes the touch disappears because the aggressor’s marketable remainder consumed it. The flag above does not establish that: it fires when the touch price moves away *or* its size more than halves, and the second clause captures cancellations as well as executions. Those are opposite mechanisms. A remainder eating the queue is mechanical. A maker *withdrawing* quotes fifty milliseconds after a hidden print is another participant revising beliefs on the information the print revealed—which is the Glosten–Milgrom mechanism operating, not an artifact of it.

ITCH separates them message by message. For a buy print the touch is the ask, where resting orders are sell limit orders, so within $(t, t + 100\text{ms}]$ at the pre-print ask price we record whether a type-4 (visible execution) message appears, a type-2/3 (cancel or delete) message appears, or both.

What happened at the touch	Share	Markout from t	From $t + 1\text{s}$
Consumption only (execution)	20.6%	+3.573 (56.9)	+0.354 (7.7)
Withdrawal only (cancellation)	28.7%	+1.842 (58.5)	+0.249 (12.0)
Both	14.6%	+3.215 (68.2)	+0.163 (4.8)
Neither (price moved otherwise)	36.1%	—	—
<i>Unswept prints</i>	—	+0.397 (24.2)	+0.381 (24.1)

Table 16: Decomposition of the sweep flag into taker consumption and maker withdrawal (470 names, 2023–2024, 235,940 name-days; shares are of swept prints; bps, NW t in parentheses). Withdrawal is involved in 43% of sweeps (28.7% alone plus 14.6% with consumption) against 35% involving consumption.

Two things follow, and they cut in opposite directions.

The displacement reading was too strong. Maker withdrawal is involved in 43% of sweeps and taker consumption in 35%; withdrawal alone is the single largest category. A substantial part of what the previous subsection attributed to mechanical displacement is therefore liquidity providers responding to the print—price discovery under any reading. The honest statement is that the majority of the measured level coincides with the touch changing, and that the touch changes for informational reasons at least as often as for mechanical ones.

But the mechanism does not survive the first second. Measured from $t + 1s$, all four groups collapse into a band of +0.16 to +0.38 bps, and the *unswept* group is the highest of them. Whatever distinguishes a consumed touch from a withdrawn one from an untouched one, it is fully expressed within a second and predicts nothing thereafter. So while the interpretation of the level changes, the forward footprint does not: it is approximately +0.4 bps regardless of what happened at the touch.

6.5 What the level actually is

Three readings of the same quantity are defensible, and we report the range rather than choose. If sweep-coincident movement is information, the level is the from- t markout on all aggressive prints. If it is displacement, the level is the unswept figure. If only withdrawal-driven sweeps count as information, it lies between. Section 5.3 carries this through to the adverse-selection share, which is the quantity that matters for interpretation, and reports it as 22%–51% of the effective half-spread accordingly. What the data do not support is quoting the top of that range unconditionally.

7 What Unconditional Markouts Overstate

Everything above is a forward markout: a measurement of where price goes from the print. A forward markout cannot distinguish response from continuation. If conditioning on aggressive same-side execution selects moments at which the midpoint is already travelling in that direction, the markout measures a move already underway. This section tests that directly and reports that it is substantially true.

7.1 The event-time profile

Table 17 measures the signed midpoint move over windows *ending* at the print as well as the forward markout.

Signed midpoint move (bps)	Outside the quote
$-30s \rightarrow t$	+1.195 (+35.6)
$-5s \rightarrow t$	+0.965 (+23.6)
$t \rightarrow +180s$ (the markout)	+1.322 (+27.3)

Table 17: Event-time profile around aggressive hidden prints (468 names, 206,235 name-days; NW t -statistics in parentheses). Moves are signed by trade direction, so a positive figure means price was already travelling *with* the trade before it arrived. Restricted to prints executing outside the prevailing quote, where the aggressor’s side cannot be misassigned. A pooled column over all aggressive prints is not reported: there the sign is assigned against the contemporaneous midpoint, so a just-fallen midpoint mechanically labels a print a buy, and the resulting pre-drift is a classification artifact.

On prints executing beyond the prevailing quote, where the aggressor’s identity cannot be mistaken, price moves *with* the trade direction before it arrives: +1.195 bps over the preceding thirty seconds ($t = 35.6$) and +0.965 over the preceding five. Over the window $[-30s, +180s]$ the signed move totals +2.52 bps, of which 47.5% occurs before the print.

7.2 Is the markout continuation of the pre-drift?

The accounting above shows that roughly half the signed move precedes the print. It does *not* show that the forward markout is continuation of that pre-drift, and the two are different claims: the first is about where the move sits in event time, the second about whether the pre-drift predicts the post-print move. We test the second directly. Within each name-day we regress the signed forward markout on the signed pre-print drift over the preceding thirty seconds,

$$m_h = a_h + b_h d + \varepsilon, \quad \text{so that} \quad \overline{m_h} = a_h + b_h \bar{d} \quad (2)$$

holds exactly within the name-day. The intercept a_h is the component of the markout orthogonal to the prior price path—the quantity a path-conditioning estimator such as the VAR is designed to isolate—and $b_h \bar{d}$ is the part attributable to continuation. Averaging both across the panel decomposes the reported markout into two pieces that sum to it by construction.

	3 min	10 min	30 min
<i>Outside-the-quote prints</i> (mean pre-drift $\bar{d} = +1.301$, $t = +37.0$)			
Markout $\overline{m_h}$	+1.379 (32.0)	+1.228 (23.4)	+1.067 (16.3)
Orthogonal to pre-drift, a_h	+1.407 (36.0)	+1.285 (24.2)	+1.154 (17.3)
Continuation, $b_h \bar{d}$	−0.028 (−1.9)	−0.057 (−3.1)	−0.088 (−3.2)
<i>All aggressive prints</i> (mean pre-drift $\bar{d} = −1.684$, $t = −55.6$)			
Markout $\overline{m_h}$	+1.153 (36.5)	+1.099 (24.8)	+1.030 (16.9)
Orthogonal to pre-drift, a_h	+1.005 (28.9)	+0.885 (21.5)	+0.756 (11.9)
Continuation, $b_h \bar{d}$	+0.148 (7.8)	+0.215 (9.7)	+0.274 (13.8)

Table 18: Decomposition (2) of the forward markout into a component orthogonal to the preceding thirty seconds of price movement and a continuation component (470 names, 2023–2024; 119,483 name-days outside-the-quote, 201,271 all-aggressive; bps, NW t in parentheses). Rows sum to the markout by construction. On the clean outside-the-quote population the continuation component is economically zero and slightly *negative*: the pre-drift does not forecast the post-print move.

The result is not what the event-time accounting led us to expect. On the outside-the-quote population—the one where the sign cannot be wrong—price does drift +1.301 bps into the print, but that drift carries essentially none of the forward markout. The continuation component is −0.028 bps at three minutes, −2% of the total and of the wrong sign, and reaches only −0.088 at thirty. The slope b_h is mildly negative throughout: prints arriving after a *larger* run-up are followed by slightly *smaller* forward moves, not larger ones. Essentially the entire markout is orthogonal to the prior path. The all-aggressive column tells the same story with the sign of $\bar{d}$ flipped by the contamination described above, and a continuation share of 13–27%.

Two conclusions follow, and they pull in opposite directions.

The specific reverse-causality mechanism is not supported. “The markout measures continuation of a move already underway” makes a testable prediction—that conditioning on the move already underway should absorb the markout—and the prediction fails. Roughly half the

signed move does precede the print, but the two halves are not the same phenomenon: the pre-drift is uninformative about what follows.

And this is not an artifact of conditioning at the wrong frequency. The decomposition above uses a single thirty-second summary of the prior path, while the VAR conditions on thirty *one-minute* lags—a richer set at a coarser frequency. Flow chasing returns at the one-to-thirty-minute scale would produce exactly the observed pattern: no continuation of thirty-second drift, yet full absorption once the minute-scale path is conditioned on. We therefore re-run the decomposition against the VAR’s own conditioning set, regressing the forward markout on the thirty preceding one-minute signed returns within each name-day. It does not absorb the markout either. The raw mean is +1.051 ($t = 47.9$); the intercept after thirty one-minute lags is +0.931 ($t = 48.8$), so the entire minute-scale path accounts for 0.121 bps, or 11.5%, with a mean within-name-day R^2 of 0.35. The bridge fails at the VAR’s frequency as well as at ours.

This does not restore a causal reading, and it does not reconcile the VAR. Ruling out one confound is not identification: flow may still be selected on information the observable price path does not reveal, and (2) conditions on a single thirty-second summary of that path rather than on its full history. We therefore retain the association language of §1.

7.3 Consequences

First, we make no causal claim. What we report is an association between aggressive hidden execution and contemporaneous price discovery, with the direction of causality unresolved.

Second—and we correct an earlier reading of our own evidence here—the pre-print drift does *not* reconcile the markout with the structural benchmark. Table 19 reports a Hasbrouck (1991) permanent–transitory VAR, re-estimated on one-minute bins with 30 lags so that the horizons of interest lie inside the model’s memory rather than beyond it, with ridge shrinkage in place of discarding explosive fits and all name-days retained rather than conditioning on stationarity. The cumulative response peaks within a minute, is barely distinguishable from zero at three, and is negative by ten. The thirty-minute cell is trim-sensitive—at a 10^4 trim its t runs between -1.9 and $+1.4$ —so the robust statement is decay to zero or below by ten minutes, and we do not lean on the thirty-minute magnitude. The natural reading—that the VAR conditions on the prior return path while the markout does not, so the gap is the pre-drift—is the one Table 18 rules out: conditioning on the prior path removes essentially none of the markout. What separates the two estimators is therefore specification, not contamination, and Table 20 identifies which feature of the specification is responsible.

Cumulative response to a 1-SD flow innovation	bps	NW t	95% CI
1 minute	+0.031	+7.80	[+0.024, +0.039]
3 minutes	+0.015	+2.27	[+0.002, +0.029]
10 minutes	−0.046	−4.17	[−0.068, −0.025]
30 minutes	−0.195	−6.57	[−0.253, −0.137]

Table 19: Re-estimated Hasbrouck (1991) VAR: one-minute bins, 30 lags (30 minutes of model memory), ridge shrinkage, all name-days retained (473 names, 121,252 name-days). The stationary-only subsample gives the same pattern (-0.049 at 10 minutes, -0.160 at 30). Magnitudes are not comparable to the markout in levels—ridge shrinkage attenuates the coefficients and the response is denominated per flow-innovation standard deviation on a one-minute clock—so we read the sign and shape, not the scale.

Two features distinguish the current specification from the earlier one that reported a permanent response: the sampling clock together with the induced memory horizon, and ridge shrinkage. Table 20 crosses them.

Specification	3 min	10 min	30 min	% stat.
10s clock, 12 lags, no ridge	+0.043 (10.0)	+0.047 (9.4)	+0.047 (9.0)	99%
10s clock, 12 lags, ridge	+0.042 (9.9)	+0.046 (9.6)	+0.048 (10.6)	99%
60s clock, 30 lags, no ridge	+0.017 (2.6)	−0.047 (−4.1)	−0.145 (−9.5)	88%
60s clock, 30 lags, ridge	+0.015 (2.4)	−0.051 (−4.8)	−0.149 (−9.4)	89%

Table 20: Cumulative VAR response to a one-SD flow innovation (bps) across the 2×2 specification grid (470 names, 2023–2024; NW t in parentheses; responses trimmed at $|\cdot| \leq 50$ bps, since non-stationary fits on the one-minute clock have heavy tails—at a 10^4 trim the three-minute and ten-minute columns are unchanged while the thirty-minute column becomes noise, t between -1.9 and $+1.4$). Rows 1–2 and 3–4 are within a rounding error of one another; rows 1 and 3 differ in sign.

The comparison is unambiguous. Ridge shrinkage does nothing: holding the clock fixed, the responses agree to the third decimal at every horizon. The clock does everything: at ten and thirty minutes the one-minute specification is negative and the ten-second specification is positive.

The reason is visible in the first row. A VAR(12) on a ten-second clock carries 120 seconds of memory, so responses requested at three, ten and thirty minutes all lie beyond the model’s horizon and return its asymptote—which is why that row reports $+0.043$, $+0.047$, $+0.047$, the same number three times. The apparent permanence of the original specification was not an estimate of behaviour at ten and thirty minutes; it was the model’s fixed point, reported at horizons it could not see. Only the one-minute clock with thirty lags places those horizons inside the estimable range, and there the response decays and turns negative.

This resolves what the specification difference is, and it removes the earlier estimate as evidence for permanence. It does not resolve the disagreement with the markouts, which remains: a path-conditioning estimator with genuine memory finds decay where the direct measurement finds none, and §7.2 has already shown that the obvious bridge between them—continuation of pre-print drift—does not carry the weight. We report the VAR as a negative result, do not cite it as corroborating permanence, and leave the disagreement open.

Third, the post-print component is not zero and is not inflated by the pre-drift: $+1.379$ bps at three minutes on the clean population, of which $+1.407$ is orthogonal to the preceding thirty seconds. Something happens after the print that does not happen before it, and it is not a continuation of what happened before it.

Establishing that it is *caused* by the print would require variation in flow unrelated to the information environment, which this design does not provide; Table 18 removes one specific alternative rather than delivering identification. We record the causal question as open, and note that it does *not* bear on the bifurcation of Section 4, which compares two contemporaneous populations rather than claiming what an execution does to prices.

8 How Much of the Footprint Is the Construction?

Everything above measures a burst formed as a run of consecutive same-side prints, where “side” is the print’s price against the contemporaneous midpoint. Sign therefore enters *formation*, not

only measurement, and that is a circularity we had not isolated: if the midpoint is drifting, prints land mechanically on one side of it and a run appears without any clustering of trader intent. The signing checks of §4.6 vary how a print is *signed*; none of them varies how a burst is *formed*. This section removes the price-conditioning in two stages and reports what is left.

8.1 Price-free formation

We hold the outcome fixed and rebuild the construction. Arm A is the published rule. Arm B forms clusters on inter-arrival timing alone—no sign condition—and signs each print against the midpoint one millisecond *before* it, which the print cannot have moved. Arm C forms clusters the same way but signs only prints executing outside the *pre-print* quote, where the aggressor’s side is unambiguous whatever the midpoint does.

	Formation / signing	3 min	15 min	30 min
A	same-side runs / contemporaneous mid	+1.482 (38.1)	+1.441 (20.5)	+1.416 (18.3)
B	time clusters / pre-print mid	+0.602 (14.9)	+0.589 (13.1)	+0.614 (13.7)
C	time clusters / outside pre-print quote	+0.029 (0.5)	−0.042 (−0.6)	−0.101 (−1.0)

Table 21: Burst-level footprint as price-conditioning is removed from the construction (470 names, 2023–2024, 235,940 name-days; bps, NW t in parentheses). Arm B retains 41% of Arm A; Arm C is statistically zero.

The footprint degrades monotonically as the construction loses sight of price. That is the signature the circularity hypothesis predicts, and it is not a power problem: Arm C rests on 4.6 million clusters.

8.2 No construction at all

Arm C changes formation *and* signing at once, and its clusters are sparse, so it cannot attribute the collapse to either. It also sits badly with §4.6, which reports +2.09 on outside-the-quote prints. We therefore drop clustering entirely and treat every hidden print as its own observation. Nothing here is formed, so nothing here can inherit formation circularity; the only remaining choice is how a print is signed.

Signing rule (per print, no clustering)	From t	From $t + 1s$
Contemporaneous midpoint (the rule used above)	+1.375 (43.9)	+0.530 (20.8)
Outside the pre-print quote only	+0.621 (12.4)	+0.097 (2.1)

Table 22: Per-print footprint under two signing rules, measured from the same base (470 names, 235,940 name-days; bps, NW t). The outside-quote figure is +0.543 at fifteen minutes and +0.495 at thirty. Only 41 of 1,604 hidden prints per day—about 2.5%—are unambiguously signable.

Three things follow, and we prefer to state them than to let the earlier numbers stand.

The footprint is real but small. Cleanly signed and unclustered, it is +0.621 bps at three minutes ($t = 12.4$), persisting at +0.495 out to thirty. It is not zero, so Arm C was too pessimistic—clustering plus the requirement of two outside-quote prints per cluster destroyed a genuine signal.

It is roughly one third of the headline. Holding the measurement base fixed and changing only the signing rule moves the per-print footprint from +1.375 to +0.621. The +2.09 of

Table 5 rests on three choices that each push the same way: contemporaneous-mid signing, price-conditioned formation, and measurement from the burst-end midpoint. We therefore report +0.62 as the conservative level and treat +2.09 as an upper bound obtained under conventions that inflate it.

Almost all of it is instantaneous. Measured from one second after the print, the cleanly-signed footprint is +0.097 ($t = 2.1$)—statistically marginal and economically nothing. This is consistent with §6.3 and §6.4, and it means the object being measured is a repricing that happens at the moment of execution, not a drift that follows it.

8.3 What is actually robust here

The stable finding across this section is not a magnitude. It is that the measured informativeness of hidden liquidity varies by a factor of three—from +2.09 to +0.62, and to a statistical zero under Arm C—according to conventions that the literature does not usually report: whether midpoint prints are signed or abstained on, whether the midpoint used is contemporaneous or lagged, whether burst formation conditions on side, and whether the clock starts at the print or a second later. Every one of those is a defensible choice. None of them is innocuous.

We regard that convention-dependence, together with the aggressive-versus-midpoint split it cannot explain away, as this paper’s contribution, and the level as provisional.

9 Scope and Limits

9.1 One venue

This is the binding limitation and we state it first. LOBSTER reconstructs the NASDAQ book alone, so every midpoint, quote and depth here is NASDAQ’s rather than the consolidated NBBO. The pre-print evidence of Section 7 makes this first-order rather than incidental: +1.195 bps of pre-drift followed by +1.322 bps of post-print move is equally consistent with adverse selection and with the NASDAQ midpoint lagging the NBBO while aggressive flow routes to hidden liquidity at prices that are already stale. The two readings are observationally equivalent in single-venue data.

Re-signing and re-marking against a consolidated quote is therefore the natural next step and, in our view, a precondition for interpreting the *magnitudes* in this paper as price discovery. We state what that work is rather than gesturing at it, since the data are standard rather than exotic: millisecond TAQ carries the NBBO, the re-run requires classifying the same type-5 prints against the consolidated quote in place of the NASDAQ quote and re-marking the same horizons, and no new estimator is involved. Every magnitude in Sections 4–7 would be re-estimated under it. We report the present results as provisional on that comparison.

The bifurcation is unaffected in kind: a contemporaneous comparison between two populations observed on the same feed does not depend on whether that feed leads or lags a consolidated one. It would, however, be *sharpened* by consolidated data in one specific way. Our at-midpoint population is defined by the NASDAQ midpoint, and a print at that midpoint may sit away from the NBBO midpoint, where it would acquire a genuine quote-rule sign instead of falling back on the tick rule—which §4.3 shows to be a weak instrument. Consolidated signing would therefore reclassify some of the at-midpoint population and replace the paper’s weakest classifier on its second-largest table.

9.2 Horizon

Every markout here is measured within a single session, at most 6.5 hours. We therefore describe the aggressive footprint as intraday-persistent rather than permanent, and test separately whether it reverts over the following month.

An earlier version of this test sorted names cross-sectionally on the day’s signed aggressive-hidden flow and measured the signed return from that day’s close to k days later, using non-overlapping formation dates. At $k = 20$ that leaves roughly 25 independent cohorts, and we reported a point estimate of -8.4 bps with $t = -1.1$ as underpowered but uncomfortable. That estimate turns out to be an artifact of the design rather than a weak signal. Because the non-overlapping sample retains only every twentieth formation date, the estimate depends on *which* phase one starts from: sweeping the starting offset across all twenty phases of the same data moves the twenty-day figure from -36.2 to $+25.5$ bps, a spread of 61.6 bps with no change to signal, weights, or estimator. The published number was one draw from that spread.

The efficient design is a calendar-time portfolio with overlapping holds. On each day the book holds the k cohorts formed on each of the previous k days at $1/k$ weight, so every trading day contributes an observation on a diversified portfolio rather than every k -th day contributing one to a sparse sort; the k -day cumulative return is k times the mean daily portfolio return, and inference runs on the daily series. Weights are the same cross-sectionally demeaned, unit-gross-exposure book used throughout.

Horizon	Cumulative (bps)	NW t	Daily (bps)
1 day	-0.69	-0.55	-0.688
2 days	-0.36	-0.32	-0.181
3 days	-0.50	-0.37	-0.166
5 days	-1.11	-0.42	-0.222
10 days	-1.08	-0.34	-0.108
20 days	-1.95	-0.40	-0.098

Table 23: Multi-day behaviour after the burst day, calendar-time portfolio with overlapping k -day holds (2023–2024 hidden panel, 474 names, 501 daily observations at every horizon; Newey–West, 10 lags). No horizon shows significant reversion and every point estimate lies within 2 bps of zero. The corresponding non-overlapping sort at twenty days ranges from -36.2 to $+25.5$ bps across starting phases of the same data; its average across phases, -1.1 bps, agrees with the calendar-time estimate here.

Under the efficient design the multi-day picture is quiet. Every point estimate lies within 2 bps of zero, none approaches significance, and the twenty-day figure is -1.95 bps against an intraday footprint of roughly $+2$. The reversion that the earlier design appeared to show does not survive using all of the data rather than a twentieth of it.

We do not overstate what this settles. The standard error at twenty days is still about 4.9 bps, so a reversion of the order of the footprint itself remains inside the confidence interval; what has been removed is the evidence *for* such a reversion, not the possibility of one. The honest statement is that the footprint is measured intraday, that nothing detectable happens to it over the following month, and that the earlier suggestion otherwise was a design artifact we have now corrected.

9.3 Universe

The name list is fixed by a liquidity screen among the tickers the LOBSTER archive stages, and that staged set is itself survivor-conditioned: names that later delisted are absent from the files we can reach rather than present as truncated series. The universe is therefore not point-in-time.

The consequence is asymmetric across our claims and we separate them. The bifurcation of Section 4 is estimated *within* name-days—aggressive against at-midpoint prints of the same name on the same day—so a missing tail of names cannot generate it. The cross-sectional *magnitudes*, by contrast, including the level of the footprint and the spread-scaling slope, are estimated on a panel missing exactly the wide-spread, distressed tail where adverse selection should be most severe, and do not carry a population interpretation. Whether the delisted names’ order flow is retrievable from the vendor is a question we have not settled.

9.4 Attribution

We observe anonymous message-level executions, not trader identities. There are no parent orders, no broker records, and no linkage to holdings. We therefore make no claim about who submits these executions, and in particular do not identify them with institutional metaorders; where that literature is invoked it is as a comparison of magnitudes. The party bearing the adverse-selection cost is whoever posted the hidden resting order, which may be a designated market maker or an institution resting size, and the data cannot distinguish them.

9.5 Interpretation

Finally, we separate the facts from the mechanism. That the footprint is a stable fraction of the quoted half-spread is a measurement. That the spread is *set* to compensate providers for it is the Glosten–Milgrom interpretation, which our evidence is consistent with but does not establish: we have not tested the causal direction between spread and adverse selection, and both could respond to a common factor. We report the stable ratio as a fact and the equilibrium mechanism as a reading of it.

10 Conclusion

Hidden liquidity on a single venue is informationally bifurcated. Executions with an identifiable aggressor are associated with a positive signed midpoint move, flat from three minutes to the close; executions at the midpoint carry a small negative one. The *level* of the first is convention-dependent—+2.09 bps as conventionally measured, +0.62 when signed unambiguously without forming a burst (§8)—and we quote the lower figure. Any classifier that force-signs midpoint prints averages the two and reports something close to zero, which is why the informativeness of hidden liquidity cannot be stated as one number. This is the self-selection equilibrium of Zhu (2014) measured inside one venue and one order type, where the confounds of a cross-venue comparison are absent.

The aggressive footprint behaves as an adverse-selection quantity, though its level depends on a classification we cannot make cleanly. It is roughly one third of the quoted half-spread and holds that share across a fivefold range of spread widths; volatility, which predicts it univariately, contributes nothing once the spread is controlled; it is between 22% and 51% of the effective half-spread on the Huang–Stoll accounting, depending on the treatment of sweep-coincident movement; it is incremental to visible order-flow imbalance; and it survives the removal of information-event

windows. Mechanical displacement is the one alternative we cannot dismiss: the majority of the measured level sits in the 30% of prints whose touch is swept within 100ms and coincides with that sweep, and at that resolution displacement and information taken through the trade are the same event.

What it is not is a tradable signal, and the reason is structural rather than incidental. A quantity that holds a fixed proportion of the spread across the cross-section is not a mispricing awaiting arbitrage; capturing it requires being the liquidity provider, and the liquidity provider is already compensated by the spread that contains it. Our own attempt to convert the footprint into a signal is consistent with this and is not reported here.

Two results limit the reading. Nearly half the signed move precedes the print on the cleanest subset available, and a properly specified VAR conditioning on the prior return path finds no lasting response. We therefore report an association rather than an impact. We are able to rule out the mechanism that would tie those two facts together—decomposing the markout against the pre-print drift leaves it essentially untouched, so it is not continuation of a move already underway—but ruling out one alternative is not identification, and the disagreement with the structural benchmark is left open rather than explained. And because the book is NASDAQ’s alone, we cannot separate price discovery from a local midpoint adjusting toward a consolidated quote that moved first. The bifurcation survives all of this; the magnitudes await a consolidated feed.

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